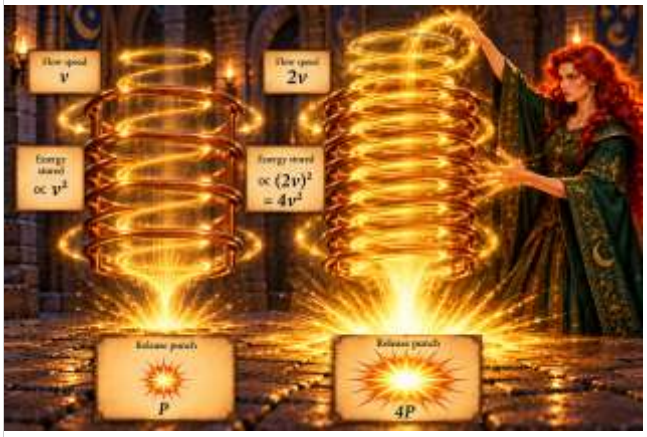
Electromagnetic Induction — Variables Reference

Inductors, induction and transformers: names, symbols, units and the formulas each quantity appears in.

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Inductance	**L**	henry	**H**	$\tau = L \div R$ $E = \frac{1}{2}LI^2$ $XL = 2\pi fL$ $\varepsilon = -L \times \Delta I \div \Delta t$ $f_0 = 1 \div (2\pi\sqrt{(LC)})$	
Magnetic flux	**Φ (phi)**	weber	**Wb**	$\Phi = BA$	
Magnetic flux density	**B**	tesla	**T**	$\Phi = BA$	
Induced EMF	**ε (epsilon)**	volt	**V**	$\varepsilon = -\Delta\varphi \div \Delta t$ $\varepsilon = -L \times \Delta I \div \Delta t$	

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Energy stored	**E**	joule	**J**	$E = \frac{1}{2}LI^2$	
Time constant	**τ (tau)**	second	**s**	$\tau = L \div R$	
Inductive reactance	**XL**	ohm	**Ω**	$XL = 2\pi fL$	
Resonant frequency	**f_0**	hertz	**Hz**	$f_0 = 1 \div (2\pi\sqrt{(LC)})$	
Change in current	**ΔI**	amp	**A**	$\varepsilon = -L \times \Delta I \div \Delta t$	
Time taken	**Δt**	second	**s**	$\varepsilon = -\Delta\varphi \div \Delta t$ $\varepsilon = -L \times \Delta I \div \Delta t$	Different from TAU!

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Coil area	**A**	square metre	**m²**	$\Phi = BA$	
Change in flux	**Δφ**	weber	**Wb**	$\varepsilon = -\Delta\varphi \div \Delta t$	Change in 
Primary voltage	**Vp**	volt	**V**	$Np \div Ns = Vp \div Vs$	
Primary turns	**Np**	(no unit — a count)	—	$Np \div Ns = Vp \div Vs$	
Secondary turns	**Ns**	(no unit — a count)	—	$Np \div Ns = Vp \div Vs$	
Secondary voltage	**Vs**	volt	**V**	$Np \div Ns = Vp \div Vs$	
Frequency	**f**	hertz	**Hz**	$X_L = 2\pi fL$ $f_0 = 1 \div (2\pi\sqrt{(LC)})$	